

# MINI-PROJECT 2 SHORT TECHNICAL REPORT

**Course:** Cross-Platform Mobile App Development (VKU)  
**Topic:** Week 6 - React Native Part 2: Navigation, State, Query, Animations  
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**Submission Date:** 30/09/2026  
**Live Demo:** <https://vku-room-booking.pages.dev/>  
**Repository:** <https://github.com/minhduy6868/mob-lab-room-booking>

## 1. PROJECT OVERVIEW

VKU Room Booking is a campus room reservation app for labs, seminar rooms, library spaces, makerspaces and smart classrooms. Week 6 upgrades the Week 5 app with typed navigation, separated client/server state, realtime cache sync, native-feeling alerts, Reanimated motion and native gestures.

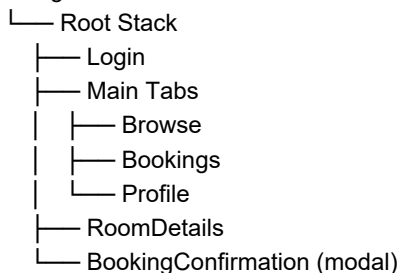
Demo account: abc@vku.udn.vn / vku@2026 (Nguyễn Văn Duy - 23IT038).

## 2. WEEK 6 IMPLEMENTATION CHECKLIST

| Requirement                  | Status   | Implementation                                                                                                   |
|------------------------------|----------|------------------------------------------------------------------------------------------------------------------|
| Root Stack + Bottom Tabs     | Complete | RootNavigator owns Login, Main tabs, RoomDetails and BookingConfirmation modal.                                  |
| Type-safe route params       | Complete | RootStackParamList defines RoomDetails { roomId, roomName } and BookingConfirmation { bookingId }.               |
| Zustand persist              | Complete | Auth session persists; booking store persists selected rooms/bookings/filter cache while KV stays authoritative. |
| TanStack Query realtime sync | Complete | Live bookings upsert into Zustand and Query cache after POST/PUT; focus/reconnect and pull refresh reconcile KV. |
| In-app alert UI              | Complete | AppAlertProvider replaces browser/native alerts with one polished notice/confirm modal.                          |
| Reanimated animations        | Complete | Room cards use FadeInDown, FadeOutUp and Layout.springify for polished list transitions.                         |
| Gesture Handler              | Complete | BookingCard supports swipe-left cancel with Gesture.Pan, shared values and runOnJS.                              |
| Code quality                 | Complete | Strict TypeScript, typed navigation, memoized RoomCard, FlatList windowing and Safe Area setup.                  |

## 3. NAVIGATION ARCHITECTURE

NavigationContainer



RoomDetails is pushed above the tab navigator so the tab bar is hidden. BookingConfirmation is presented as a modal digital pass after POST /api/bookings succeeds.

## 4. STATE AND DATA FLOW

Zustand handles client state: auth session, selected date/filter, selected modal room, selected QR pass and cached bookings. TanStack Query handles server state: room catalog and live booking records. Cloudflare Pages Functions with KV remain the source of truth for conflict validation, cancellation and check-in.

The app no longer polls KV on a timer. Successful POST/PUT responses return the changed booking, then the client upserts that record into Zustand and the TanStack Query cache immediately.

Browse card -> RoomDetails -> Booking sheet -> POST /api/bookings

- > server validates conflict rules

- > store/query cache upsert the returned booking record

- > BookingConfirmation modal opens with bookingId

## 5. MOTION, GESTURE AND DEMO SCRIPT

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Room cards animate into the list with Reanimated. Booking cards expose a red cancel action under the card; a left swipe past the threshold triggers the same confirm-safe cancellation path as the visible cancel button.

Demo: login -> tap a room card -> RoomDetails -> book an available slot -> BookingConfirmation pass -> open My Bookings and see the new booking without interval reload -> pull refresh only when reconciling cloud state -> swipe left to cancel -> try an occupied/overlapping slot and verify the 409 message.

## 6. NOTES

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NPM reported existing dependency audit findings after installing Reanimated. No automatic npm audit fix was applied because it may introduce breaking dependency upgrades outside the Week 6 assignment scope.